

Exact Residual-Query Memory and Smooth-Loss Hierarchy Limits

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Abstract

This working paper establishes two mathematical boundaries for a project on information, retained summaries, and neural growth. First, exact universal empirical residual queries for centered quadratic predictors are equivalent to a finite collection of label, second-order, and fourth-order moments. With $p = \binom{d+1}{2}$, $q = \binom{d+3}{4}$, and fixed unweighted sample size $n \geq p + q$, the minimum dimension of a continuous real-valued summary is exactly $2p + q$, or $p + q$ additional coordinates when the label moment is already supplied. At $d = 12$ these counts are 1,521 and 1,443. The result concerns every dataset and every query; it is not an approximate-memory, finite-word, or generalization lower bound. Second, a single smooth strictly convex binary generator with finite endpoint slopes has unbounded optimal hierarchy price in excess Bayes risk, even with four equally likely source states. Along the constructed family indexed by j , deterministic and stochastic prices are asymptotic to $j/2$ and $j/4$, respectively. The corresponding total-risk prices tend to one. A classical covariance-sketch guarantee gives an explicit approximate query baseline. An appendix separates representation information, forgetting, Bayes value, and fitted risk, including a zero-output neural birth that gains representation information while preserving prediction risk. Classical ingredients and prior project results are identified; no priority or general learning guarantee is claimed.

1 Questions, context, and the limits of the claims

The preceding project studied finite representation hierarchies, including logarithmic-loss obstructions and curvature comparisons [7], and growth in a trainable quadratic network. Its Phase 6 experiment asked whether a retained historical moment improves final risk under a declared work budget, rather than merely supplying a promising birth direction [9]. That experiment and the earlier theory notes already distinguished population moment formulas from exact empirical residuals [8, 9]. The present paper addresses two mathematical questions left separate from those empirical comparisons: what must a summary retain to answer every empirical residual query, and whether finite endpoint slopes suffice to prevent an unbounded excess-risk price of a compatible hierarchy.

For observations (x_i, y_i) , put $H_i = x_i x_i^\top - I_d$ and let the centered quadratic predictor be $q_B(x) + c = \langle B, x x^\top - I_d \rangle + c$. Phase 6 proposes directions using $S_y - B$, where $S_y = (2n)^{-1} \sum_i y_i H_i$. The exact empirical matrix residual is instead

$$G(B, c) = \frac{1}{2n} \sum_i \{y_i - \langle B, H_i \rangle - c\} H_i = S_y - T(B) - \frac{c}{2} \bar{H}, \quad T(B) = \frac{1}{2n} \sum_i \langle B, H_i \rangle H_i. \quad (1)$$

For standard Gaussian input and fixed sample-independent B , $\mathbb{E}T(B) = B$ and $\mathbb{E}\bar{H} = 0$. Replacing empirical quantities by those population identities does not reproduce a realized query, especially after B has been chosen from the same history. The memory result characterizes the stronger

Result	Required scope	Conclusion and boundary
Exact residual memory	Every real-valued size- n dataset; $n \geq p+q$; continuous encoder; arbitrary decoder	Exactly $2p + q$ real coordinates, with all decoder-accessible state counted. No physical-byte or Gaussian average-case claim.
Finite-bit packing	Simultaneous error on a specified finite query family; a realizable local moment cube	A local high-accuracy bit bound. Its radius is not shown dimension independent.
Approximate queries	Spectral covariance-sketch error and a bounded query norm	Uniform deterministic query error by an existing sketch guarantee; no population-risk bound.
Smooth-loss hierarchy	One fixed generator; four equally weighted source states; deterministic nesting or stochastic degradation	Excess prices grow as $j/2$ and $j/4$. The displayed total-risk prices tend to one.

Table 1: Claim domains. An ideal real coordinate is not a finite-precision word. Exact universal queries impose a stronger requirement than supporting one prescribed optimization trajectory.

requirement of exact answers for *all* B, c on *all* datasets. It neither claims that the implemented population-based proposal ought to be exact nor assigns its empirical performance to a missing moment.

The two principal results have different domains and should not be combined into an assertion that neural growth must fail or succeed. The first is a representation theorem for empirical computations. The second is an existence theorem for optimal nested partitions under one fixed proper-loss generator. Neither theorem describes finite-budget training of neural parameters. Table 1 records the assumptions needed to read the conclusions.

The main tools have established antecedents. Empirical residuals are normal-equation computations, finite polynomial moments are classical representations, and invariance of domain gives dimensional obstructions [3, 10, 16]. Continuity restrictions and pathological single-real encodings already play an explicit role in the theory of set representations [19]. The finite hierarchy problem uses Bregman centroid identities and the established proper-score representation [2, 6, 17]. An upper bound on a generator’s first derivative is substantially weaker than a uniform positive lower curvature bound or a global quadratic comparison [1]. We use those distinctions to state a boundary result, not a complete classification or a claim of first discovery.

2 Exact empirical residual-query memory

For the remainder of this section let

$$p = \binom{d+1}{2}, \quad q = \binom{d+3}{4}, \quad D_0 = p + q, \quad D = 2p + q.$$

These count symmetric second moments, symmetric fourth moments, and the label–quadratic cross moment. The continuous-dimension result concerns a fixed known sample count $n \geq D_0$.

2.1 Definition, query equivalence and a matching summary

Fix $d, n \geq 1$ and a known sample count n . Data are an ordered tuple $z = ((x_i, y_i))_{i=1}^n \in (\mathbb{R}^d \times \mathbb{R})^n$. Permutation invariance is not required of a competing summary; allowing order only strengthens

the lower bound. Set

$$H(x) = xx^\top - I_d, \quad q_B(x) = \text{tr}(BH(x)),$$

$$G_z(B, c) = \frac{1}{2n} \sum_{i=1}^n (y_i - q_B(x_i) - c)H(x_i), \quad B \in \text{Sym}_d, \quad c \in \mathbb{R}.$$

A summary $S_n : z \mapsto \mathbb{R}^m$ is *universally query-sufficient* if there is a decoder Γ_n such that

$$\forall z \forall B \in \text{Sym}_d \forall c \in \mathbb{R} : \quad \Gamma_n(S_n(z), B, c) = G_z(B, c). \quad (\text{U})$$

No regularity of Γ_n is needed for the continuous lower bound. The decoder has no additional data-dependent side information unless explicitly stated; all retained model/optimizer/replay information that it accesses must be counted in the summary. A known sample count is free in this fixed- n model. A streaming implementation stores the count too.

Write $\langle a \rangle_z = n^{-1} \sum_i a(x_i, y_i)$ and define

$$S_y = \frac{1}{2} \langle yH \rangle_z, \quad M_{2,ij} = \langle x_i x_j \rangle_z, \quad M_{4,ijkl} = \langle x_i x_j x_k x_l \rangle_z.$$

The subscripts on M_2, M_4 are coordinate indices rather than sample indices. Store only distinct symmetric coordinates: $i \leq j$ for M_2 and $i \leq j \leq k \leq l$ for M_4 . Let

$$F_n(z) = M(z) = (S_y, M_2, M_4) \in \mathbb{R}^D.$$

The order-four tensor has q distinct entries, not $p(p+1)/2$ independent entries. Products of centered quadratic features have polynomial relations, and known total mass is not an additional unknown coordinate.

Theorem 2.1 (Canonical query information). *For any two datasets z, z' of the same known size, the following are equivalent:*

1. $G_z(B, c) = G_{z'}(B, c)$ for every symmetric B and every c ;
2. $(S_y, \overline{H}, T)_z = (S_y, \overline{H}, T)_{z'}$, where $\overline{H} = \langle H \rangle$ and $T(B) = \frac{1}{2} \langle \langle B, H \rangle H \rangle$;
3. $M(z) = M(z')$.

Consequently M answers every query, and every sufficient summary permits recovery of M . This is minimality for a deterministic query family, not a statement of statistical sufficiency for an unknown parameter.

Proof. Direct expansion gives

$$G_z(B, c) = S_y - T(B) - \frac{1}{2} c \overline{H}. \quad (\text{M1})$$

Thus the value at $(0, 0)$ determines S_y , its difference from $(0, 1)$ determines $\overline{H}$, and its differences from queries at a basis of Sym_d determine T . Conversely these coefficients determine every query.

For $k \leq l$, put $F^{kk} = e_k e_k^\top$ and $F^{kl} = (e_k e_l^\top + e_l e_k^\top)/2$ if $k < l$, so that $\langle F^{kl}, H \rangle = H_{kl}$. Define

$$K_{ij,kl} = \langle H_{ij} H_{kl} \rangle = 2[T(F^{kl})]_{ij}.$$

The reconstruction formulas are

$$M_{2,ij} = \overline{H}_{ij} + \delta_{ij}, \quad (\text{M2})$$

$$M_{4,ijkl} = K_{ij,kl} + \delta_{ij} M_{2,kl} + \delta_{kl} M_{2,ij} - \delta_{ij} \delta_{kl}. \quad (\text{M3})$$

Conversely (M3) reconstructs K from M_2, M_4 , and hence reconstructs T . All index permutations are determined by tensor symmetry. If two histories have the same sufficient summary, (U) makes all their queries agree, so the reconstructed M agrees too. Therefore a map h on the summary image satisfies $M = h \circ S_n$. $\square$

The $p + 2$ fixed queries

$$\mathcal{Q} = \{(0, 0), (0, 1)\} \cup \{(F^{kl}, 0) : 1 \leq k \leq l \leq d\} \quad (\text{M4})$$

already recover all coefficients. This reduces any exact universal adaptive-query claim to a finite nonadaptive query family. Once M is stored, correctness is pathwise and holds for arbitrarily data-dependent choices of B, c ; independence or generalization assumptions are unnecessary. If only one prespecified algorithm trajectory must be supported, (U) is a stronger requirement and the lower bound need not apply.

The summary is additive: store sums of $yH/2$, all degree-two input monomials, and all degree-four input monomials, then divide by n . This uses D ideal real coordinates and a count, independently of stream length. It is an algebraic exact-arithmetic representation; ordinary floating-point accumulation has rounding error. Nothing here identifies the real-coordinate count with a physical byte count.

2.2 Continuous summaries on fixed unweighted samples

Let $f(x) \in \mathbb{R}^{D_0}$ list all degree-two and degree-four monomials. The functions $1, f_1, \dots, f_{D_0}$ are linearly independent. The additional p functions $yH_{ij}(x)/2$ are independent modulo this input-only span. Indeed, the coefficient of y in a linear relation would say $x^\top Ax - \text{tr}(A) \equiv 0$, forcing $A = 0$; independence of different input monomials then finishes the argument. Thus the affine span of the single-example canonical feature vector is all of $\mathbb{R}^D$.

Lemma 2.2 (A full-dimensional unweighted moment image). *For every fixed $n \geq D_0$, there is a dataset z_0 at which the polynomial moment map $F_n : (\mathbb{R}^{d+1})^n \rightarrow \mathbb{R}^D$ has differential of rank D . Consequently its image contains a nonempty open subset of $\mathbb{R}^D$, and there is a continuous local section $s : U \rightarrow (\mathbb{R}^{d+1})^n$ on some nonempty open $U \subset \mathbb{R}^D$ such that $F_n(s(u)) = u$.*

Proof. First consider $A_n(x_1, \dots, x_n) = n^{-1} \sum_i f(x_i)$. The vectors $\partial_a f(x)$, over all $x \in \mathbb{R}^d$ and coordinates a , span $\mathbb{R}^{D_0}$. Otherwise a nonzero v annihilates them all, so every partial derivative of the polynomial $v^\top f$ vanishes. That polynomial is constant on $\mathbb{R}^d$, contradicting independence of $1, f_1, \dots, f_{D_0}$.

Select D_0 such derivative vectors forming a basis, and use their locations as D_0 sample points. Their selected derivative columns in DA_{D_0} are independent. Pad the configuration when $n > D_0$. Hence some D_0 -minor of DA_n is a nonzero polynomial in the sample coordinates. The configurations where it is nonzero form an open dense set $\mathcal{U}$.

Independence of the p polynomials H_{ij} implies that there are p points for which the $p \times p$ evaluation matrix has rank p : if their value vectors never spanned $\mathbb{R}^p$, a nontrivial linear combination of the functions would vanish everywhere. Since $n \geq D_0 \geq p$, the $p \times n$ matrix with columns $H(x_i)$ in symmetric coordinates has rank p on another open dense set $\mathcal{V}$. Both density assertions follow because a nonzero real polynomial cannot vanish on an open set. The finite intersection $\mathcal{U} \cap \mathcal{V}$ is nonempty.

Choose an input configuration there and set every label to zero. At this dataset, the derivative of S_y with respect to input coordinates is zero, while its label-derivative columns are $H(x_i)/(2n)$, of rank p . The derivative of (M_2, M_4) with respect to inputs has rank D_0 and its label derivatives

vanish. Thus the full differential has rank $p + D_0 = D$. Choose D independent scalar input/label coordinate directions and freeze the others. The inverse function theorem applied to the resulting square map supplies U and its smooth, hence continuous, local section. $\square$

Theorem 2.3 (Tight continuous dimension for sufficiently many samples). *Fix $n \geq D_0$. Any continuous universally query-sufficient summary $S_n : (\mathbb{R}^{d+1})^n \rightarrow \mathbb{R}^m$ must have $m \geq D$. This bound is attained by M . If S_y is supplied separately and an additional continuous summary A_n has values in $\mathbb{R}^m$, exact universal decoding from (S_y, A_n) requires and permits $m \geq D_0$.*

Proof. On the local section in the lemma, $S_n \circ s : U \rightarrow \mathbb{R}^m$ is continuous. It is injective: equal summary values imply equal query coefficients by the first theorem, and $F_n(s(u)) = u$. If $m < D$, compose this injection with the usual coordinate embedding $\mathbb{R}^m \hookrightarrow \mathbb{R}^D$. Brouwer’s invariance of domain says its image is open in $\mathbb{R}^D$, but it lies in an m -dimensional coordinate subspace with empty interior. This is a contradiction.

For separate S_y , the combined map (S_y, A_n) is a continuous summary of dimension $p + m$, so $p + m \geq D$. Storing M_2, M_4 attains $m = D_0$. Neither proof assumes continuity of the decoder. $\square$

Finite sample qualification. The sufficient threshold $n \geq D_0$ is not claimed to be the smallest threshold. For general n , let

$$r_n = \max_z \text{rank } DF_n(z).$$

The same independent-minor argument and a local parameter slice imply that every continuous universal summary has dimension at least r_n . Moreover $r_n \leq \min\{D, n(d+1)\}$ and storing the ordered raw data uses $n(d+1)$ continuous coordinates. Thus a D lower bound cannot hold when $n(d+1) < D$. At $n = 1$, for example, storing (x, y) uses only $d+1$ coordinates. These bounds do not determine the exact minimum for every small n ; a sharper small- n classification can involve the geometry of finite-atomic polynomial moment varieties.

Quantifiers and model restrictions. The theorem assumes correctness at every dataset in the stated unrestricted domain. It is not automatically a lower bound for data on a fixed noiseless teacher manifold, for a discrete alphabet, or for almost-sure correctness with an arbitrary discontinuous decoder. Even topological full support of a sampling distribution does not turn an almost-sure factorization through a discontinuous decoder into a pointwise factorization. If both encoder and the finitely many decoders in (M4) are continuous, equality almost surely under a full-support law extends by continuity to every dataset, so the theorem then applies. A fixed shared random seed also reduces to this deterministic theorem only if that seed’s map has universal correctness.

2.3 Linear measurements and why additivity alone is insufficient

The domain in this section is all finitely supported probability measures μ on $\mathbb{R}^{d+1}$, with arbitrary nonnegative real weights, not merely equal weights. Let $\phi_1, \dots, \phi_m$ be any real-valued functions and store their linear measurements $L(\mu) = (\int \phi_j d\mu)_j$. Here “linear” means linear in the measure, not linear in x or y .

Proposition 2.4 (Exact linear-measurement criterion). *For a target function f , the scalar $\int f d\mu$ is determined by $L(\mu)$ for every finitely supported probability measure if and only if*

$$f \in \text{span}_{\mathbb{R}}\{1, \phi_1, \dots, \phi_m\}. \tag{M5}$$

Therefore universal exact quadratic residual queries require $m \geq D$ in this measurement model, regardless of decoder regularity. The canonical measurements attain equality.

Proof. The “if” implication follows by integrating a linear representation, using known total mass one. For the converse, put $v(z) = (1, \phi_1(z), \dots, \phi_m(z))$. If every finite real relation $\sum_i a_i v(z_i) = 0$ also implied $\sum_i a_i f(z_i) = 0$, then $\ell(\sum_i a_i v(z_i)) = \sum_i a_i f(z_i)$ would be a well-defined linear functional on the span of the $v(z)$. Extend it to $\mathbb{R}^{m+1}$; then $f(z) = \ell(v(z))$, establishing (M5). If (M5) fails, there is instead a finite relation with $\sum_i a_i = 0$ and $\sum_i a_i f(z_i) \neq 0$. Split the coefficients into positive and negative parts. Their common positive mass a normalizes them into two finitely supported probability measures μ_+, μ_- with equal L but unequal integrals of f . No decoder can distinguish them.

Apply this criterion to the D canonical feature functions. Their classes modulo constants are linearly independent, as proved above, so the span of m measurement classes must have dimension at least D . $\square$

An equivalent finite-support kernel statement is $\ker L \cap \ker(\text{mass}) \subseteq \ker M$: every signed measure annihilated by the stored measurements and mass must annihilate all query coefficients. This is ordinary linear algebra. It is not a separate streaming or information-theoretic principle.

Two models that must not be conflated. For equal-weight samples of a fixed size, arbitrary discontinuous additive features do not satisfy the lower bound in the proposition. Choose a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$ and assign each possible record $z \in \mathbb{R}^{d+1}$ to a distinct basis element b_z ; both sets have cardinality continuum. The scalar sum $\sum_i b_{z_i}$ uniquely specifies every multiplicity in the finite multiset, by rational linear independence. Thus one additive exact real, together with the known count, stores all empirical query information. This encoding is not a continuous finite-precision algorithm. Its existence shows why one cannot infer a lower bound from “additive” plus a nonlinear decoder alone. Arbitrary-real-weight mixtures defeat this example because real, rather than rational, linear relations are then allowed. Likewise, without any encoder regularity, the cardinality equality $|\mathbb{R}^D| = |\mathbb{R}|$ permits storing the whole moment vector in one ideal real. Even measurability alone is not a replacement for continuity: standard digit encodings can be made Borel measurable. This discontinuous-sum loophole is established prior theory: see Wagstaff et al., Theorem 3.2 and Supplement B.2[19].

If a decoder is required to be affine in the stored feature averages, the span criterion and D lower bound follow already from testing single records (or a dataset consisting of n copies of one record). This is a different, stronger restriction on decoding.

2.4 A finite-bit lower bound with explicit error metric

Let $\|A\|_{\max} = \max_{ij} |A_{ij}|$. Define simultaneous query error on the fixed finite family (M4) by

$$e_z = \max_{(B,c) \in \mathcal{Q}} \|\widehat{G}_z(B, c) - G_z(B, c)\|_{\max}. \quad (\text{M6})$$

This metric is specified because an unqualified phrase such as “ ε -accurate residual queries” does not define a memory problem. All matrices F^{kl} have Frobenius norm at most one, so the query family uses bounded coefficients.

Lemma 2.5 (Stable recovery from the finite query family). *If $e_z \leq \varepsilon$, the formulas in the first theorem reconstruct a vector $\widehat{M}$ satisfying $\|\widehat{M} - M(z)\|_{\infty} \leq 12\varepsilon$.*

Proof. The errors in $S_y = G(0, 0)$, $\overline{H} = 2(G(0, 0) - G(0, 1))$, and $K_{ij,kl} = 2(G(0, 0) - G(F^{kl}, 0))_{ij}$ are at most ε , 4ε , and 4ε , respectively. Equation (M2) has error at most 4ε . Equation (M3) adds at most two such errors to the error of K , giving 12ε . To select one value for each fourth-order

coordinate, fix any deterministic pair grouping in (M3); approximate reconstructions need not otherwise respect redundant tensor identities. $\square$

Theorem 2.6 (Local worst-case bit bound). *Fix $d \geq 1$ and $n \geq D_0$. There exist a moment vector m_0 , $r > 0$, and a continuous section s on a neighborhood of $C = m_0 + [-r, r]^D$, with $F_n(s(m)) = m$. Fix the compact family of size- n unweighted datasets*

$$\mathcal{K} = s(C).$$

Any summary with at most 2^b states and any decoder satisfying (M6) with $e_z \leq \varepsilon$ for every $z \in \mathcal{K}$ must obey, for $\varepsilon > 0$,

$$b \geq D \log_2 \left(1 + \left\lfloor \frac{2r}{25\varepsilon} \right\rfloor \right). \quad (\text{M7})$$

If S_y is supplied exactly as uncharged side information, the number of additional bits has the same lower bound with D_0 replacing D . No finite number of states can be exact on all of $\mathcal{K}$. These claims require correctness only on this compact family, not on all real-valued datasets.

Proof. The local section gives an open realizable moment set. Choose a closed cube C inside it; its continuous image $\mathcal{K} = s(C)$ is compact. In each coordinate place $L = 1 + \lfloor 2r/(25\varepsilon) \rfloor$ equally spaced grid values with spacing 25ε ; choose one realizing dataset via the section for each of its $N = L^D$ grid points. If two grid datasets had the same stored state, the decoder would return the same fixed query outputs for both. Recovering M from these outputs would approximate both moment vectors within 12ε , so their distance would be at most 24ε , contradicting the grid separation. Hence there are at least N states, proving (M7).

For side information, fix the first p coordinates of the cube (the label moment) at their center, and vary only the remaining D_0 coordinates. All datasets now have identical side information, and the same proof applies. For zero error, even a line segment inside either cube has infinitely many distinct exact query vectors, so finite-state universal exact recovery is impossible. $\square$

The compact-family restriction makes a fixed positive absolute error meaningful. If one instead required the same guarantee over all unbounded real-valued datasets, already $G_z(0, 0) = S_y$ has unbounded range, so no finite-state code could have a finite uniform absolute error. Such a global impossibility would not be a useful quantitative precision bound.

This is a packing/pigeonhole theorem, not an invocation of Fano's inequality without a probability model. It has an optional precise randomized version. Draw the dataset uniformly from the constructed N -point packing. A randomized b -bit encoding/decoding scheme with shared randomness independent of the dataset and simultaneous success probability at least $1 - \delta$ on each dataset satisfies

$$b \geq \log_2 N + \log_2(1 - \delta). \quad (\text{M8})$$

Condition on all random coins: each state can be successful for at most one packed dataset, so at most 2^b of the N datasets succeed; average this inequality over the coins. This also covers private coins by revealing all of them as free shared randomness. A per-query success guarantee is different from simultaneous success for (M4), and requires its own union-bound adjustment.

For fixed d, n and the resulting local scale r , (M7) gives $D \log_2(1/\varepsilon) - O_{d,n,r}(1)$ as $\varepsilon \downarrow 0$. The proof does not give a dimension-uniform lower bound on r , a Gaussian-prior rate-distortion function, or a practical fixed-precision word count. The same compact family has a matching-order elementary upper bound. For $\eta = \varepsilon/3$, use a grid on each canonical coordinate interval of length $2r$ with

$J = 1 + \lceil 2r/\eta \rceil$ values and spacing at most η . Encoding the nearest grid vector has coordinate error at most η and uses no more than

$$b_{\text{upper}} = \left\lceil D \log_2 \left(1 + \left\lceil \frac{6r}{\varepsilon} \right\rceil \right) \right\rceil \quad (\text{M9})$$

bits. In (M1)–(M3), the errors in S_y , $\overline{H}$, and K are at most η , η , and 3η ; hence every query in (M4) has error at most $3\eta \leq \varepsilon$. The grid vector need not itself be a realizable empirical moment vector for this decoder to work. If S_y is supplied exactly, quantize only the D_0 input moment coordinates and replace D by D_0 in (M9). Thus on the specified compact family, the lower and upper bounds have matching $D \log_2(1/\varepsilon)$ leading order for fixed d, n, r .

In particular, neither the continuous exact theorem nor this local high-accuracy packing result proves that useful approximate residual queries require $\Omega(d^4)$ storage at a fixed practical tolerance. Classical covariance sketches such as Frequent Directions explicitly trade storage against covariance approximation error[5]; their applicability and bounds for the lifted quadratic features require a separate calculation. The local radius r in (M7) must not be silently treated as dimension independent.

2.5 Concrete collisions and the Phase6 boundary

The missing fourth moment is operational. For $d = 1, n = 2$ and both labels zero, take dataset A with $x = (-1, 1)$ and dataset B with $x = (0, \sqrt{2})$. Both have $S_y = 0$ and $M_2 = 1$ (equivalently $\overline{H} = 0$). Dataset A has $\langle H^2 \rangle = 0$, whereas dataset B has $\langle H^2 \rangle = 1$. At $B = 1, c = 0$, their exact query values are 0 and $-1/2$. Thus even S_y together with the empirical second moment cannot answer all residual queries. Replicating each record equally gives the same collision at any even sample size.

Implementation inspected. The Phase 6 source `code/frontier.py`[9] updates

$$\text{moment} += \frac{1}{2} \sum_{\text{new } i} y_i (x_i x_i^\top - I), \quad \text{matrix} = \text{moment}/n - B.$$

Thus the proposal uses $S_y - B$. The exact historical empirical quantity is (M1). For Gaussian inputs, $\mathbb{E}T(B) = B$ for fixed sample-independent B and $\mathbb{E}\overline{H} = 0$, but these are population identities. They do not turn $S_y - B$ into the realized empirical query. The Phase 5 and Phase 6 theory files[8, 9] already state this distinction correctly; this note supplies qualified minimality results that those files deliberately did not claim.

With the Phase6 value $d = 12$,

$$p = 78, \quad q = 1365, \quad D_0 = 1443, \quad D = 1521.$$

The sufficient sample-size threshold in the continuous lower bound is therefore $n \geq 1443$. With 1024 observations per arrival it applies from the second arrival if the query target is the whole historical empirical dataset. This observation is conditional on (U) and on what information the decoder can access. It is not a claim that the implemented approximate controller violates a memory constraint. A dense symmetric $p \times p$ Gram matrix plus $\overline{H}$ would use $p(p+1)/2 + p = 3159$ stored scalar slots, whereas the input polynomial moments use only 1443; adding S_y gives 3237 versus 1521. These compare representations, not bit lower bounds or runtime. Known constants and exact identities explain the redundancy.

Changing the requested output changes the target dimension. The matrix query $G(B, c)$ does not contain $\langle y \rangle$ or $\langle y^2 \rangle$. To reproduce the intercept gradient as well, store $\langle y \rangle$, one more independent coordinate:

$$\langle y - q_B - c \rangle = \langle y \rangle - \langle B, \bar{H} \rangle - c.$$

To reproduce the entire squared empirical loss, add $\langle y^2 \rangle$ too. The corresponding unrestricted moment dimensions are $D + 1$ and $D + 2$. Exact matrix gradient queries therefore do not alone reproduce factor Adam with its chosen minibatches, replay selection, coordinatewise states, and clipping.

Removing the intercept is dimension dependent. One must not simply subtract p coordinates when restricting queries to $c = 0$. For $d \geq 3$, the centered Gram moments already determine $\bar{H}$. The pointwise rank-one identities of $H + I$ imply

$$\begin{aligned} \langle H_{ii} + H_{jj} \rangle &= \langle H_{ij}^2 - H_{ii}H_{jj} \rangle - 1 & (i \neq j), \\ \langle H_{jk} \rangle &= \langle H_{ij}H_{ik} - H_{ii}H_{jk} \rangle & (i, j, k \text{ distinct}). \end{aligned}$$

Three pairwise diagonal sums recover each diagonal entry, and the second identity recovers each off-diagonal entry. Thus for $d \geq 3$ the dimension remains D even without c . For $d = 1$, the only input query coefficient is $\langle H^2 \rangle$, so the total affine moment dimension is two including S_y . For $d = 2$, the six distinct Gram polynomials are independent modulo constants: their degree-four leading terms force all coefficients to vanish except possibly a multiple of $H_{11}H_{22} - H_{12}^2$, and its degree-two part is nonzero. Hence the total is $6 + p = 9$, compared with $D = 11$ when all intercepts are queried. These are affine moment-space counts; the finite- n qualifications still apply.

2.6 Prior theory and defensible contribution

- **Classical least squares.** In the feature vector $h(x) = \text{vec}_{\text{Sym}}(xx^\top - I)$, equation (M1) is the normal-equation residual $X^\top y - X^\top X \beta$, with the intercept cross term. MIT’s official notes give the ordinary normal equations and their gradient derivation[10]. The matrix query identity, Gram storage, and regression interpretation are established theory.
- **Statistical sufficiency is a different quantifier.** The official Berkeley notes define sufficiency through the conditional distribution of data given a statistic, relative to a specified statistical model, and minimal sufficiency through recovery from any sufficient statistic[18]. This note instead fixes a family of deterministic empirical queries. For a fixed design Gaussian regression model, the design is conditioned upon and its Gram matrix need not be charged as stored unknown information. This is another reason not to borrow statistical minimal-sufficiency claims as memory bounds here.
- **Polynomial moments and kernel means.** Sriperumbudur et al. study mean embeddings of measures and conditions for full distributional injectivity[14]. Taking the finite feature map to be our canonical polynomials makes M its empirical kernel mean. It distinguishes this query family but does not determine the full data distribution. The finite-dimensional feature interpretation, and the distinction between query equivalence and a characteristic kernel, are established theory.
- **Moment compression is classical.** Bayer and Teichmann prove a version of Tchakaloff’s theorem: finitely many weighted evaluation nodes can reproduce polynomial integrals through a fixed degree[3]. This concerns existence of positive cubature representations, not a lower bound on physical memory. It reinforces that finite polynomial moment representations are old, and does not remove the coordinate or bit cost of describing nodes and weights.

- **Topology is classical.** The only topological input is Brouwer’s invariance of domain; Tao’s author-written exposition gives the theorem, its no-injection corollary, and a proof[16]. The dimension lower bound is its application to a locally full-rank empirical moment map.
- **Set-summary dimensional obstructions are prior work.** Wagstaff et al. make continuity versus discontinuous single-real encodings explicit and prove continuous latent-dimension obstructions for sum representations of set functions[19]. Their Theorem 4.1 proof uses an injection from an open ordered-set domain into the latent space. Our specialization concerns a fixed finite quadratic query family, and permits any continuous summary, including nonadditive and order-dependent ones. The dimension obstruction and the need to exclude pathological encodings are not new principles.

The defensible project contribution is the precise specialization: the $q + 2p$ count after polynomial dependencies and labels are charged; the $q + p$ additional cost given S_y ; a proof for sufficiently large fixed equal-weight samples rather than only arbitrary weighted measures; and the finite-query accuracy metric that yields a bit packing bound. This package is presented as a precise boundary for the project’s historical summary, with classical ingredients credited and no claim of priority. An algorithmic or learning-theoretic claim needs a separate result.

3 A classical approximate-query baseline

The exact dimension theorem does not imply that approximate residual queries need fourth-order storage. Here is an explicit transfer from an existing matrix-sketch guarantee. This is a corollary, not a new sketching algorithm.

Let $\text{svec} : \text{Sym}_d \rightarrow \mathbb{R}^p$ be the Frobenius isometry which stores diagonal entries unchanged and off-diagonal entries multiplied by $\sqrt{2}$. Form $A \in \mathbb{R}^{n \times p}$ with row i equal to $\text{svec}(H(x_i))^\top$. Store S_y and $\overline{H}$ exactly, and let $C \in \mathbb{R}^{\ell \times p}$ satisfy

$$\|A^\top A - C^\top C\|_2 \leq \eta.$$

Define

$$\widehat{G}(B, c) = S_y - \text{svec}^{-1} \left(\frac{C^\top C}{2n} \text{svec}(B) \right) - \frac{c}{2} \overline{H}.$$

Proposition 3.1 (Uniform deterministic query transfer). *For every $R \geq 0$, simultaneously for every $\|B\|_F \leq R$ and $c \in \mathbb{R}$,*

$$\|G(B, c) - \widehat{G}(B, c)\|_F \leq \frac{R\eta}{2n}.$$

The guarantee therefore remains true for queries chosen adaptively from the summary or the entire data, without an independence assumption.

Proof. The isometry and the empirical query formula give

$$\text{svec}(G - \widehat{G}) = -\frac{A^\top A - C^\top C}{2n} \text{svec}(B).$$

The operator norm inequality proves the result, pointwise and uniformly. $\square$

Frequent Directions gives, for every integer $0 \leq k < \ell \leq p$, $\eta \leq \|A - A_k\|_F^2 / (\ell - k)$, where A_k is a best rank- k approximation [5, Theorem 1.1]. Thus the query error is at most $R\|A - A_k\|_F^2 / [2n(\ell - k)]$ using $\ell p + 2p$ stored real coordinates for $C, S_y, \overline{H}$, plus a count and implementation workspace.

This count excludes floating-point rounding and is not a claim of optimality for the polynomially constrained feature rows. The cited theorem concerns arbitrary matrices; any approximate lower bound from that paper cannot automatically be restricted to rows $\text{svec}(xx^\top - I)$.

This baseline has no distribution assumption, but its tail bound can be large. It neither bounds out-of-sample risk nor proves that the selected growth direction is useful. A next research target is a specified memory–accuracy tradeoff on these polynomial feature rows, with a separate population-error and optimization analysis. No sketching implementation or new training experiment was run in Phase 7.

4 Equal source masses do not control hierarchy price for smooth losses

For a convex generator ϕ and a finite source with binary posteriors η_x , define

$$D_\phi(\mathcal{P}) = \sum_x p_x \phi(\eta_x) - \sum_{C \in \mathcal{P}} p_C \phi(\bar{\eta}_C), \quad \bar{\eta}_C = p_C^{-1} \sum_{x \in C} p_x \eta_x.$$

Let $D_{\phi,k}^*$ be its flat k -cell minimum. A deterministic four-state hierarchy is a nested pair $(\mathcal{H}_2, \mathcal{H}_3)$ with exactly two and three cells; the one-cell and singleton levels can always be appended. Its minimax excess price is

$$\rho_{\phi,\text{det}} = \min_{\mathcal{H}} \max_{k=2,3} D_\phi(\mathcal{H}_k) / D_{\phi,k}^*.$$

For a channel $X \rightarrow Z$, set $D_\phi(Z) = \mathbb{E}\phi(\eta_X) - \mathbb{E}\phi(\mathbb{E}[\eta_X | Z])$. The stochastic price takes the infimum of the same maximum over $Y - X - Z_3 - Z_2$ with output budgets three and two. No public randomization variable is supplied to the predictor. All denominators below are positive.

Theorem 4.1 (A fixed smooth generator with an equal-mass obstruction). *There exists a single symmetric function $\phi \in C^\infty([0, 1])$ with $0 \leq \phi'' \leq 1$, $\phi'' > 0$ on $(0, 1)$, and finite endpoint derivatives, for which the following holds. For every integer $j \geq 16$ there is a full-support four-state binary-target law with $p_A = p_C = p_D = p_B = 1/4$ such that*

$$\rho_{\phi,\text{det},j} = \frac{j^2 - j + 8}{2j + 4} = \frac{j}{2} - \frac{3}{2} + \frac{7}{j + 2}, \quad \rho_{\phi,\text{stoch},j} \sim \frac{j}{4}. \quad (\text{F1})$$

The generator is strictly convex, while all its derivatives of order at least two vanish at zero and one. Its associated strictly proper binary losses can be smooth and bounded in $[0, 1]$.

Thus a positive source-mass bound does not by itself control the excess hierarchy price uniformly across fixed smooth bounded-slope generators. The logarithmic source-mass bound relies on properties of logarithmic loss. The theorem does not assert an unbounded total-risk price.

4.1 A smooth four-point gadget with exact costs

Fix an even nonnegative C^∞ density κ supported in $[-1, 1]$, with integral one, and write $\kappa_\eta(u) = \eta^{-1} \kappa(u/\eta)$. Define a smoothed hinge

$$H_\eta(s) = \int_{\mathbb{R}} (s - u)_+ \kappa_\eta(u) du.$$

It is convex and smooth, with $H''_\eta = \kappa_\eta$. Symmetry gives zero mean for the kernel, so $H_\eta(s) = 0$ for $s \leq -\eta$ and $H_\eta(s) = s$ for $s \geq \eta$. For $0 < \varepsilon \leq 1/16$, take $\eta = \varepsilon/4$ and put

$$F_\varepsilon(z) = H_\eta(-z - \varepsilon) + H_\eta(z - 1 - \varepsilon) + 4\varepsilon^2(z - 1/2)^2. \quad (\text{F2})$$

It is symmetric about $1/2$ and has $F''_\varepsilon \geq 8\varepsilon^2$. For this intermediate calculation use the four real locations

$$(z_A, z_C, z_D, z_B) = (-1, 0, 1, 2), \quad p_x = 1/4.$$

These locations will later be transformed to valid binary posteriors.

Lemma 4.2 (Exact gadget costs and deterministic optimum). *Let a, b, r, q, t, u denote, respectively, a near pair AC or DB , a far pair AD or CB , the middle pair CD , the outer pair AB , a triple with one outer point, and a triple with both outer points. Their Jensen costs for (F2) are*

$$\begin{aligned} a &= \varepsilon/4 + \varepsilon^2/2, & r &= \varepsilon^2/2, & b &= t = (1 - \varepsilon)/4 + 2\varepsilon^2, \\ q &= (1 - \varepsilon)/2 + 9\varepsilon^2/2, & u &= (1 - \varepsilon)/2 + 14\varepsilon^2/3. \end{aligned} \quad (\text{F3})$$

The unique flat optima are $AC \mid DB$ and $A \mid CD \mid B$, with costs $2a$ and r . The deterministic price is

$$\min\{\alpha, \beta\} = \beta, \quad \alpha = \frac{a}{r} = \frac{1}{2\varepsilon} + 1, \quad \beta = \frac{t}{2a} = \frac{1 - \varepsilon + 8\varepsilon^2}{2\varepsilon + 4\varepsilon^2}. \quad (\text{F4})$$

Every deterministic risk-ratio vector is coordinatewise at least one of $(1, \alpha)$ and $(\beta, 1)$, and both vectors are attained.

Proof. The possible means of nonempty subsets of the four equally weighted locations belong to

$$\{-1, -1/2, 0, 1/3, 1/2, 2/3, 1, 3/2, 2\}.$$

Each lies outside the smoothing bands of width η around $-\varepsilon$ and $1 + \varepsilon$. At all points used in any partition cost, the hinge part of (F2) consequently agrees exactly with

$$V_\varepsilon(z) = (-z - \varepsilon)_+ + (z - 1 - \varepsilon)_+.$$

Its values at the four inputs are $(1 - \varepsilon, 0, 0, 1 - \varepsilon)$. The near-pair mean is $-1/2$ or $3/2$, and its hinge cost is $(1 - \varepsilon)/4 - (1/2)(1/2 - \varepsilon) = \varepsilon/4$. The middle-pair mean is $1/2$ and its hinge cost is zero. The far-pair and one-outer-point triple means are zero or one; their hinge costs are $(1 - \varepsilon)/4$. The outer-pair mean is $1/2$, and the two-outer-point triple mean is $1/3$ or $2/3$; their hinge costs are $(1 - \varepsilon)/2$.

The Jensen cost of $(z - 1/2)^2$ equals weighted within-cell variance. For these six cell types its values are respectively $1/8, 1/2, 1/8, 9/8, 1/2, 7/6$, in the order a, b, r, q, t, u . Multiplication by $4\varepsilon^2$ proves (F3).

There are seven two-cell partitions, with costs $2a, 2b, q + r, t, t, u, u$, and six three-cell partitions, with costs a, a, b, b, r, q . For $\varepsilon \leq 1/16$ one has $r < a < b$, $q > a$, $2a < t$, and $t < q + r, u$. These strict inequalities identify the unique flat minima.

If the fine partition merges CD , its possible coarse costs are $t, t, q + r$, so $(\beta, 1)$ is its best ratio vector. If the fine partition merges a near pair, the coarse optimum $AC \mid DB$ is available, attaining $(1, \alpha)$. Any other fine pair has cost at least a , so its vector is dominated by $(1, \alpha)$. This proves the Pareto assertion and the minimum formula. Finally

$$\alpha - \beta = \frac{5\varepsilon - 4\varepsilon^2}{2\varepsilon + 4\varepsilon^2} > 0,$$

proving the exact value β . □

4.2 Stochastic refinement of the gadget

Lemma 4.3 (A sharp leading stochastic price). *For the gadget (F2), under the same two output budgets,*

$$1 + \frac{(\alpha - 1)(\beta - 1)}{\alpha + \beta - 2} \leq \rho_{\varepsilon, \text{stoch}} \leq \frac{1}{4\varepsilon} + \frac{4}{3}. \quad (\text{F5})$$

In particular $\rho_{\varepsilon, \text{stoch}} \sim 1/(4\varepsilon)$.

Proof. For any distribution on deterministic hierarchies, the Pareto assertion reduces minimization of the maximum expected ratio to a mixture of $(1, \alpha)$ and $(\beta, 1)$. Equalizing its two coordinates gives the left side of (F5).

To apply this to arbitrary stochastic chains, represent the fine encoder and its coarsener by independent random function tables, independent of the source. Conditional on the tables U , the two representations are deterministic and nested. Convexity gives

$$D_{F_\varepsilon}(Z_k) \geq \mathbb{E}_U D_{F_\varepsilon}(Z_k \mid U).$$

Indeed, revealing U makes the conditional mean more variable, so Jensen's inequality increases the expectation of F_ε being subtracted from the fixed full-information term. For each fixed table, any partitions with fewer than the respective budgets can be refined to exactly three and two cells while retaining nesting: first split the fine cells to three, keeping each inside its original coarse cell. Retain the coarse partition if it already has two cells; if it has only one, group the three fine cells into two nonempty coarse cells. Refinement decreases Jensen cost. Hence a distribution on exact deterministic hierarchies has expected coordinate costs no greater than those of the stochastic chain. This proves the lower bound without giving the predictor a free randomization label.

For the upper bound, use fine outputs $0, m, 1$ with

$$A \mapsto 0, \quad C \mapsto m, \quad D \mapsto \tfrac{1}{2}m + \tfrac{1}{2}1, \quad B \mapsto 1,$$

and coarsen $0, m$ to one output. Here the displayed mixture for D means an independent random choice between the two indicated outputs. At the fine level, the middle cell has mass $3/8$ and location mean $1/3$; the right cell has mass $3/8$ and mean $5/3$. At the coarse level, the left cell has mass $5/8$ and mean $-1/5$. All these means are outside the smoothing bands when $\varepsilon \leq 1/16$. Direct evaluation gives

$$D_{F_\varepsilon}(Z_3) = \varepsilon/8 + (2/3)\varepsilon^2, \quad D_{F_\varepsilon}(Z_2) = 1/8 + \varepsilon/2 + (26/15)\varepsilon^2. \quad (\text{F6})$$

For detail, the fine hinge cost is $\varepsilon/8$, and the sum of fine weighted variances is $1/12 + 1/12 = 1/6$. The coarse hinge cost is $(1/8 + 3\varepsilon/8) + \varepsilon/8$; its sum of variances is $7/20 + 1/12 = 13/30$. Multiplying variances by $4\varepsilon^2$ proves (F6). The fine ratio equals $1/(4\varepsilon) + 4/3$. The coarse cost minus this ratio times $2a$ is $-5\varepsilon/12 + 2\varepsilon^2/5 < 0$. This proves the upper bound. Since both α and β are asymptotic to $1/(2\varepsilon)$, the lower bound has the same leading coefficient, completing the proof. $\square$

4.3 Embedding all gadgets into one smooth generator

Proof of Theorem 4.1. For each integer $j \geq 16$, let

$$a_j = 2^{-j-2}, \quad l_j = 2^{-j-3}, \quad I_j = (a_j, a_j + l_j), \quad \lambda_j = e^{-j^2}.$$

These intervals are pairwise disjoint and accumulate only at zero. Choose a fixed smooth cutoff $\chi : \mathbb{R} \rightarrow [0, 1]$ equal to one on $[1/8, 7/8]$ and zero outside $(1/16, 15/16)$. Use the symmetric positive baseline

$$b(x) = e^{-1/[x(1-x)]} \quad (0 < x < 1), \quad b(0) = b(1) = 0.$$

For $x \in I_j$, put $t = (x - a_j)/l_j$ and define

$$g(x) = (1 - \chi(t))b(x) + \chi(t)\frac{16\lambda_j}{l_j^2}F_{1/j}''(4t - 3/2). \quad (\text{F7})$$

On the reflected intervals set $g(x) = g(1 - x)$, and elsewhere let $g = b$. Every component is positive at interior points, because $F_{1/j}'' \geq 8/j^2$ and $b > 0$. The cutoff vanishes on neighborhoods of interval boundaries, so the definitions match smoothly there.

For each fixed derivative order m , the derivatives of the kernel give $\|F_{1/j}^{(m+2)}\|_\infty \leq C_m j^{m+1} + 8/j^2$. The constant second derivative term need only be included for $m = 0$; the displayed larger bound is valid in every case. Consequently the m th derivative of the second summand in (F7) is bounded by

$$C_m \lambda_j l_j^{-m-2} (1 + j^{m+1}) \leq C_m e^{-j^2 + C_m j}. \quad (\text{F8})$$

The baseline and cutoff term has corresponding bound $C_m e^{-c2^j + C_m j}$: derivatives of b are b times rational functions with poles at the endpoints, while derivatives of the rescaled fixed cutoff contribute only powers of l_j^{-1} . Both estimates vanish faster than a_j^N for every fixed m, N . Away from the patches, the baseline has the same flatness at zero. Thus each derivative of g tends to zero faster than any power of x as $x \downarrow 0$. Defining all endpoint derivatives to be zero is consistent with differentiation, since $g^{(m)}(x)/x \rightarrow 0$; induction proves $g \in C^\infty([0, 1])$. Reflection gives the same conclusion at one.

Set $M = \max_{[0,1]} g > 0$ and define

$$\phi'' = g/M, \quad \phi(1/2) = \phi'(1/2) = 0.$$

This function is symmetric, smooth on the closed interval, and strictly convex because its second derivative is strictly positive in the interior. It has finite endpoint slopes and all derivatives of order at least two vanish at the endpoints. By normalization $0 \leq \phi'' \leq 1$.

On $J_j = [a_j + l_j/8, a_j + 7l_j/8]$, (F7) implies that

$$\phi(x) = \frac{\lambda_j}{M} F_{1/j} \left(4 \frac{x - a_j}{l_j} - \frac{3}{2} \right) + A_j x + B_j \quad (\text{F9})$$

for suitable affine constants. Now put equal source masses $1/4$ at the four binary posteriors

$$(\eta_A, \eta_C, \eta_D, \eta_B) = a_j + l_j(1/8, 3/8, 5/8, 7/8). \quad (\text{F10})$$

They are distinct, strictly between zero and one, and yield a full-support joint law. Under the coordinate transformation in (F9) they correspond to $(-1, 0, 1, 2)$. Every deterministic cell mean and every stochastic encoded posterior stays in J_j . Affine cancellation therefore proves the exact identities

$$D_\phi(\mathcal{P}) = \frac{\lambda_j}{M} D_{F_{1/j}}(\mathcal{P}), \quad D_\phi(Z) = \frac{\lambda_j}{M} D_{F_{1/j}}(Z) \quad (\text{F11})$$

for all partitions and all encoders. In particular the flat denominators are $(\lambda_j/M)2a > 0$ and $(\lambda_j/M)r > 0$. Lemma 4.2 with $\varepsilon = 1/j$ gives the exact deterministic expression (F1), and Lemma 4.3 gives the stochastic asymptotic.

To obtain the stated bounded proper loss, write $h(q) = \phi(0) - \phi(q)$ and set

$$\ell(q, 1) = h(q) + (1 - q)h'(q), \quad \ell(q, 0) = h(q) - qh'(q).$$

Their expected regret is the Bregman divergence generated by ϕ , so strict convexity proves strict propriety. Endpoint supporting lines make the losses nonnegative. They are smooth, with derivatives $\ell(q, 1)' = -(1 - q)\phi''(q)$ and $\ell(q, 0)' = q\phi''(q)$. Their largest values equal $-\phi'(0) = \phi'(1) = \int_0^{1/2} \phi''(t)dt \leq 1/2$. They hence lie in $[0, 1]$. $\square$

Total-risk limitation. The construction deliberately establishes an excess-risk boundary. For the displayed proper loss let $c = -\phi'(0) > 0$. Since $h(q) = cq + o(q)$ as $q \downarrow 0$ and all posteriors (F10) lie between a_j and $3a_j/2$, their full-information Bayes risk B_j is at least $ca_j/2$ eventually. On $[-1, 2]$, the convex nonnegative gadget $F_{1/j}$ is bounded above by its equal endpoint values $1 - 1/j + 9/j^2 < 2$. Equation (F11) therefore gives $D_\phi(Z) \leq 2\lambda_j/M$ for every encoder. All total-risk ratios to the respective flat total-risk optima are at most

$$1 + \frac{4\lambda_j}{Mca_j} = 1 + o(1).$$

Thus total-risk hierarchy prices for these very instances tend to one. The result does not show large absolute excess loss: the flat distortions and all copied costs are extremely small.

Scope and attribution. Smooth cutoffs, proper-score representations, and Jensen/Bregman centroid identities are classical; see [2, 6, 17]. Uniform curvature comparisons impose stronger assumptions than finite endpoint slopes; see [1]. The specific ingredients assembled here are the exact equal-mass hinge-gadget Pareto costs, its stochastic leading coefficient, and its embedding into one fixed smooth generator. Priority is not asserted; the accompanying primary-source audit distinguishes this statement from greedy Bregman clustering and from uniform curvature comparisons. This argument does not determine the universal sharp logarithmic-loss coefficient, the all-capacity price for larger sources, or a classification of all generators.

5 Interpretation and reproducibility

The exact memory theorem identifies what is lost when a historical summary preserves only a label-quadratic cross moment. It also identifies the assumptions that make a dimension lower bound valid. The encoder must be continuous for the continuous theorem, and every data-dependent state available to the decoder counts toward its dimension. The linear-measurement version uses arbitrary weighted measures or affine decoding; it must not be transferred to arbitrary discontinuous additive encoders on equal-weight samples. The local finite-bit bound uses a stated error norm and a nonquantified local radius. None of these exact or high-accuracy results establishes $\Omega(d^4)$ practical storage at a fixed useful tolerance.

The sketch corollary makes the approximation boundary constructive: a covariance approximation of lifted quadratic features supplies simultaneously valid approximate residuals for bounded query matrices. Whether this is useful depends on the lifted matrix's spectral tail, the desired accuracy, and the implementation cost. It remains separate from estimation of population risk, selection of a birth direction, and optimization of the enlarged network. No new training experiment or sketch implementation accompanies this theoretical increment.

The smooth-generator construction shows that a loss can have bounded values, finite slopes, and interior strict convexity while its excess hierarchy price is unbounded. The critical qualification is the absence of a uniform positive curvature lower bound. Its source weights need not become rare: all four remain $1/4$. Instead, the generator contains increasingly weak curvature patterns at smaller scales. The absolute distortions vanish, and the common Bayes baseline dominates the total risk. Thus an unbounded excess ratio need not describe a large absolute prediction penalty or an unbounded total-risk ratio.

Appendix A addresses a third, supporting distinction. Even for logarithmic loss, a change of representation has a forgetting term unless the representations form a refinement. Information gain refers to an unrestricted Bayes decoder; an unchanged output function has unchanged realized risk.

A zero-output birth can nevertheless add informative hidden features. Finite optimization, decoder restriction, and population estimation require their own analysis. These classical identities do not convert either principal result into an autonomous-growth principle.

The artifact preserves the research drafts, the imported manuscript sections, source citations, independent review notes, and exact or high-precision algebraic checks. The memory checks reconstruct queries and explicit collisions and verify small-dimensional ranks using rational arithmetic. Finite-cost enumeration checks the local partition calculations; it does not replace the smooth-gluing proof or prove an infinite asymptotic limit numerically. The build script compiles only the copied manuscript sources and bibliography using the existing local L^AT_EX installation. Previous project phases and the research authors' original files are not edited by the manuscript build. Internal project citations identify provenance; they are not externally published or independently certified results.

A What information gained by a representation does and does not buy

All logarithms are natural. Representations and the target are random variables on standard Borel spaces; conditional laws exist. The target is finite except where real regression is stated explicitly. Population risks are evaluated on a fresh example. For learned encoders, all statements can be applied conditional on a fixed training history, provided the fresh example has the stipulated conditional law. Conditioning does not assert that the encoder is independent of its training examples.

Write $B_\ell(Z) = \inf_g \mathbb{E}\ell(Y, g(Z))$, where the infimum runs over all measurable decoders with the appropriate action space. A deterministic refinement has $Z = \pi(Z')$ almost surely. More generally, the conclusions stated for refinements hold when $Y - Z' - Z$ is a Markov chain under the specified coupling. Appending a feature is a refinement; retraining and replacing an entire hidden layer need not be one. The results in this section are classical identities, elementary consequences, and explicit counterexamples. Proper-loss information and Bregman interpretations are established in [12, Theorems 8 and 10] and [6, Theorem 1 and Section 2.2].

Proposition A.1 (Information gain includes a possible forgetting term). *For finite Y , logarithmic loss gives $B_{\log}(Z) = H(Y | Z)$. For any pair of representations on a common probability space,*

$$B_{\log}(Z) - B_{\log}(Z') = I(Y; Z' | Z) - I(Y; Z | Z').$$

For a refinement, the second term vanishes, and the Bayes improvement is exactly $I(Y; Z' | Z)$.

Proof. Conditional expected log loss at $Z = z$ equals the entropy of the true conditional distribution plus its KL divergence to the prediction. The divergence is nonnegative and is zero at the true distribution. Next expand both conditional mutual informations, using $H(Y | Z, Z')$ as their common subtracted term. Finally the Markov condition makes $I(Y; Z | Z') = 0$. $\square$

Thus positive conditional mutual information concerns an optimal unrestricted decoder; it is not a decrease promised for a trained decoder. Nor does adding and then reoptimizing neurons automatically preserve the information-refinement condition.

Proposition A.2 (Squared and classification risk under refinement). *Suppose $Y - Z' - Z$ is Markov. If Y is real with $\mathbb{E}Y^2 < \infty$, put $m = \mathbb{E}[Y | Z]$ and $m' = \mathbb{E}[Y | Z']$. Then*

$$\Delta_{\text{sq}} := B_{\text{sq}}(Z) - B_{\text{sq}}(Z') = \mathbb{E}(m' - m)^2.$$

If additionally $Y \in [a, b]$ almost surely and $J = I(Y; Z' | Z)$, then

$$0 \leq \Delta_{\text{sq}} \leq \frac{1}{2}(b - a)^2 J.$$

For finite-label classification with zero-one loss,

$$0 \leq \Delta_{01} := B_{01}(Z) - B_{01}(Z') \leq \sqrt{J/2}.$$

For $Y \in \{0, 1\}$, both constants are asymptotically sharp among refinements. Here the Brier loss means $(Y - p)^2$, not the sum over both class coordinates. If the binary coarse posterior satisfies $m \in [\alpha, 1 - \alpha]$ almost surely, $0 < \alpha \leq 1/2$, then also

$$J \leq \frac{\Delta_{\text{sq}}}{\alpha(1 - \alpha)}.$$

The last bound is a convenient reverse bound, not a sharp constant claim.

Proof. Conditional expectation minimizes squared loss. The tower identity and the Markov condition give $\mathbb{E}[m' | Z] = m$, hence $\mathbb{E}[m'm] = \mathbb{E}m^2$. Subtracting the two Bayes risks gives $\mathbb{E}(m')^2 - \mathbb{E}m^2 = \mathbb{E}(m' - m)^2$.

Write $P' = P_{Y|Z'}$ and $P = P_{Y|Z}$. Under the Markov condition, $J = \mathbb{E}\text{KL}(P' \| P)$. The mean difference for a variable in $[a, b]$ is at most $(b - a)\text{TV}(P', P)$. Pinsker's inequality, in nats, states $\text{TV}(P', P)^2 \leq \text{KL}(P' \| P)/2$. Integration proves the squared-risk bound. The conditional best classification accuracy is $\max_y P(y)$. Since $\max_y P'(y) - \max_y P(y) \leq \text{TV}(P', P)$, integration, Cauchy–Schwarz, and Pinsker give the upper bound. The nonnegative sign follows from conditional Jensen, since $P = \mathbb{E}[P' | Z]$ and the maximum is convex.

For sharpness let Z be constant and let $m' = 1/2 \pm \epsilon$ with equal probabilities. Then $\Delta_{\text{sq}} = \epsilon^2$, $\Delta_{01} = \epsilon$, and $J = \log 2 - h(1/2 + \epsilon) = 2\epsilon^2 + O(\epsilon^4)$, where $h(p) = -p \log p - (1 - p) \log(1 - p)$. Finally $\log u \leq u - 1$ gives $\text{kl}(q, p) \leq (q - p)^2/[p(1 - p)]$ for $0 < p < 1$. Apply this conditionally and integrate. $\square$

The bounded-response implication is an instance of the classical information-to-conditional-mean inequality; see the earlier absolute-mean version in [15, Lemma 4.4] and the explicit sharp squared-mean formulation in [11, Corollary 7.11]. Pinsker and reverse-divergence comparisons are established tools [13, Eq. (1), Section VI]. These statements are not new neuron-growth guarantees.

Proposition A.3 (Function preservation and representation preservation differ). *If $Z' = F(Z)$ and $Z = G(Z')$ almost surely for measurable F, G , then the two representations generate the same sigma field and $I(Y; Z') = I(Y; Z)$, $I(Y; Z' | Z) = 0$. Exact neuron duplication before symmetry-breaking perturbations has this property for the complete hidden representation.*

Nevertheless, a function-preserving neural birth can have strictly positive representation information gain. Fix $0 < \delta < 1/2$, let X be uniform on $\{-1, 1\}$, and set

$$\mathbb{P}(Y = 1 | X = 1) = 1 - \delta, \quad \mathbb{P}(Y = 1 | X = -1) = \delta.$$

Start with constant representation $Z = 1$ and prediction $f = 1/2$. Append the ReLU feature $r(X) = \max\{X, 0\}$ with output coefficient zero. The new representation is $Z' = (1, r(X))$, while the new prediction is still $f' = 1/2 + 0r(X) = 1/2$. Then

$$I(Y; Z' | Z) = \log 2 - h(\delta) > 0, \quad \Delta_{\text{sq}} = (1/2 - \delta)^2.$$

The actual prediction risk at birth is unchanged for every loss for which it is defined. In particular its squared risk is $1/4$, its log risk is $\log 2$, and its classification error under the same fixed tie rule is $1/2$, both before and after birth.

Proof. Mutual measurability gives equality of sigma fields, hence equality of all conditional laws. In the neural example $r(X)$ determines X . The marginal label is balanced, with entropy $\log 2$, and its conditional entropy given X is $h(\delta)$. Its conditional mean is $\delta + (1 - 2\delta)r(X)$, at distance $1/2 - \delta$ from $1/2$. Propositions A.1–A.2 give the asserted Bayes gains. The realized predictions coincide pointwise, which proves equality of their realized risks directly. $\square$

Function-preserving duplication is established in Net2Net [4, Section 2.3]; the distinction here is between the unchanged output function and what an unrestricted decoder could recover from the enlarged hidden vector. In the displayed example even an affine decoder can express the new Bayes predictor, but the birth operation has not fitted its coefficients.

Proposition A.4 (Four explicit failures of information-only criteria). *The following claims hold, each under a deterministic refinement.*

1. *Activation entropy can be arbitrarily large while task information and every Bayes-risk improvement are zero. Let U be uniform on $\{1, \dots, M\}$, independent of $Y \sim \text{Ber}(1/2)$. Take $Z = 1$ and $Z' = \text{ReLU}(U) = U$. Then $H(Z') = \log M$, $I(Y; Z' | Z) = 0$, and the conditional target law is unchanged.*
2. *Positive label information need not improve classification. Let $Z = 1$, and let Z' be equiprobable with binary label posteriors $1/10$ and $3/10$. Strict concavity gives $J = h(1/5) - \{h(1/10) + h(3/10)\}/2 > 0$, but both Bayes classification errors are $1/5$, since the Bayes action stays zero. This rules out a universally positive classification-gain lower bound at every positive information level; it does not rule out all bounds at larger information levels.*
3. *Without restricting the response or the decision loss, even a fixed positive information gain need not reduce squared Bayes risk. Let $U \sim \text{Ber}(1/2)$ and independent S be uniform on $\{-1, 1\}$. Set $Y = S(1 + U)$, $Z = 1$, and $Z' = U$. Then $U = |Y| - 1$, so $J = \log 2$, but both conditional means are zero and both squared Bayes risks equal $\mathbb{E}Y^2 = 5/2$.*
4. *Fixed structural information cost does not imply useful risk reduction. For independent equiprobable $S \in \{-1, 1\}$ and uniform $U \in \{1, \dots, M\}$, let $X = (S, U)$, $Z = 1$, $Z' = X$, and $Y | X \sim \text{Ber}(1/2 + \epsilon S)$ with $0 < \epsilon < 1/2$. Then*

$$I(X; Z' | Z) = \log(2M), \quad J = \log 2 - h(1/2 + \epsilon), \quad \Delta_{\text{sq}} = \epsilon^2, \quad \Delta_{01} = \epsilon.$$

All three Bayes gains tend to zero as $\epsilon \downarrow 0$ while the structural information cost is fixed and can be arbitrarily large.

Proof. Independence proves the first item. The second follows by averaging the two conditional errors and entropies. The third follows from independence and the zero mean of S . In the fourth, Z' reveals X , so its input information is $H(X) = \log(2M)$. The label is balanced, the conditional entropy is $h(1/2 + \epsilon)$, the posterior mean is $1/2 + \epsilon S$, and the fine classification error is $1/2 - \epsilon$. These observations give all displayed formulas. $\square$

Even for binary labels there is no distribution-free positive constant c with $\Delta_{\text{sq}} \geq cJ$ for all refinements. Indeed let $X = Y \sim \text{Ber}(p)$, $Z = 1$, $Z' = X$. Then $\Delta_{\text{sq}} = p(1 - p)$ and $J = h(p)$, whose ratio tends to zero as $p \downarrow 0$. This is only a failure of a uniform linear comparison; it is not a claim that no nonlinear comparison exists.

Proposition A.5 (The exact gap between Bayes value and fitted risk). *For $t \in \{0, 1\}$, fix representation Z_t , an admissible decoder class $\mathcal{G}_t$, and its fitted member g_t . Assume all risks and infima below are finite. Let $R_t(g) = \mathbb{E}\ell(Y, g(Z_t))$, $B_t = B_\ell(Z_t)$, and define*

$$A_t = \inf_{g \in \mathcal{G}_t} R_t(g) - B_t \geq 0, \quad F_t = R_t(g_t) - \inf_{g \in \mathcal{G}_t} R_t(g) \geq 0.$$

Then

$$R_0(g_0) - R_1(g_1) = (B_0 - B_1) + (A_0 - A_1) + (F_0 - F_1).$$

If $\hat{R}_t$ is an empirical criterion, an exact further split is

$$F_t = O_t + E_t, \quad O_t = \hat{R}_t(g_t) - \inf_{g \in \mathcal{G}_t} \hat{R}_t(g) \geq 0,$$

$$E_t = R_t(g_t) - \hat{R}_t(g_t) + \inf_{g \in \mathcal{G}_t} \hat{R}_t(g) - \inf_{g \in \mathcal{G}_t} R_t(g).$$

Here E_t can have either sign. If $U_t = \sup_{g \in \mathcal{G}_t} |R_t(g) - \hat{R}_t(g)| < \infty$, then $|E_t| \leq 2U_t$.

Proof. Add and subtract each infimum to obtain the identities. Uniform deviation bounds both the fitted function's risk difference and the difference of the two infima by U_t in absolute value. $\square$

A_t is representation-decoder approximation error; F_t is the realized population fitting gap, not a pure optimization error. Only O_t is the empirical optimization residual in the displayed split. The algebra holds for data-dependent encoders and classes after they are fixed, but obtaining a small probabilistic bound for U_t then requires appropriate uniform control or fresh data. The identities do not supply one. A larger realizable function class can lower its infimum while finite optimization fails to lower the fitted risk.

Proposition A.6 (A total information budget requires nested retention). *Let $Z_0, \dots, Z_T$ be deterministic successive refinements of X . Then*

$$\sum_{t=1}^T I(Y; Z_t \mid Z_{t-1}) = I(Y; Z_T \mid Z_0) = B_{\log}(Z_0) - B_{\log}(Z_T) \leq I(Y; X \mid Z_0).$$

If all information quantities are finite and $C = \sum_{t=1}^T I(X; Z_t \mid Z_{t-1})$, then also

$$C = I(X; Z_T \mid Z_0), \quad B_{\log}(Z_0) - B_{\log}(Z_T) \leq C.$$

Write $M = \min\{I(Y; X \mid Z_0), C\}$. For bounded real $Y \in [a, b]$, the endpoint squared Bayes improvement is at most $(b - a)^2 M/2$. The corresponding finite-label classification bound is $\sqrt{M/2}$.

Proof. Each earlier representation is a function of every later one. Apply the mutual-information chain rule and telescope the conditional entropies of the finite target. Apply the same chain rule to X (via mutual information, without requiring finite differential entropies). Conditional data processing gives $I(Y; Z_T \mid Z_0) \leq I(Y; X \mid Z_0)$ and $I(Y; Z_T \mid Z_0) \leq I(X; Z_T \mid Z_0)$. Proposition A.2 applied once to the endpoint pair proves the remaining bounds. $\square$

The same chain-rule conclusions hold for an explicitly specified joint Markov degradation chain $Y - X - Z_T - \dots - Z_0$. They need not hold for arbitrary successive fitted representations. For example, take a balanced binary $Y = X$ and alternate $Z_t = X$ at odd t and a constant at even t , starting with constant Z_0 . Every odd transition contributes $\log 2$ to $I(Y; Z_t \mid Z_{t-1})$; every even transition contributes zero. Their sum grows without bound as the number of cycles grows, although

the final representation after any full cycle has no label information. The omitted reverse conditional information accounts exactly for each forgetting step. A cumulative transcript restores nesting, but describes the information of that retained transcript, not automatically that of the deployed network. None of these chain-rule facts controls optimizer descent or charges finite-precision storage, feature-search computation, or parameter updates.

For a learned sequence, a valid conditional version fixes one common training history H containing the randomness that specifies every encoder, and uses a fresh evaluation sample independent of H . Apply the preceding propositions under each conditional law given H , then average. Conditioning separately on a growing H_t at each stage without accounting for the newly learned information is not the same argument. In particular the randomness specifying the next encoder must be conditioned on, or represented explicitly as part of the observation, before applying a fixed-encoder Bayes comparison.

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